The Geopolitics of Oil:

Assessing the Macroeconomic Impact of the 2026 Gulf Crisis

Panel Data Analysis: Pooled OLS, Fixed Effects & Random Effects

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Panel: 12 Countries × 42 Months (Jan 2023 - Jun 2026) | N = 504

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Abstract

This paper investigates the macroeconomic transmission of global oil-price shocks and financial market volatility into domestic inflation and industrial output using an unbalanced panel of 12 countries spanning January 2023 through June 2026. The sample encompasses four structurally distinct country categories: Advanced Net-Importers (Germany, France, Italy, Japan), Advanced Net-Exporters (United States, Canada), Emerging Net-Importers (Turkey, Poland, India, Thailand), and Emerging Net-Exporters (Saudi Arabia, Russia). Pooled OLS, Fixed Effects (Within), and Random Effects (Swamy-Arora) panel regressions are estimated for two dependent variables: the monthly log-difference of the Consumer Price Index (Δln CPI) and the monthly log-difference of the Industrial Production Index (Δln IPI). Our principal empirical contribution is diagnostic rather than confirmatory: the complete collapse of the Swamy-Arora individual-specific variance component to zero ($\theta = 0$), the failure of the Breusch-Pagan LM test to detect meaningful panel effects ($p = 0.1552$ for CPI; $p = 0.2732$ for IPI), and the Hausman test's inability to distinguish between estimators ($p = 0.9086$ for CPI; $p = 0.1004$ for IPI) constitute compelling evidence that indiscriminate global pooling suppresses genuine structural heterogeneity. We interpret these findings through the lens of aggregation bias: the stagflationary pass-through observed in oil-importing economies is systematically offset in the pooled aggregate by the windfall dynamics of oil exporters, neutralising the very panel variation that would otherwise make macro-econometric inference tractable. Effective and credible identification of shock transmission channels requires structurally homogeneous sub-panels or, ultimately, country-specific time-series frameworks.

1. Introduction and Theoretical Framework

The global economy in the post-pandemic era has been characterised by an unusually compressed sequence of supply-side disruptions, geopolitical realignments, and monetary policy regime shifts. Among the most consequential of these disruptions has been the persistent volatility of international crude oil prices, amplified by the conflict in Eastern Europe beginning in early 2022, the subsequent reconfiguration of global energy trade flows, and renewed tensions in the Gulf region during late 2025 and early 2026. These developments have reignited a longstanding debate in empirical macroeconomics: through what channels, and with what magnitude, do exogenous oil price shocks propagate into domestic price levels and industrial activity across countries that are structurally differentiated by their net energy trade positions, exchange rate regimes, and states of economic development?

The theoretical architecture linking international commodity price shocks to domestic macroeconomic outcomes is well established, though empirically contested at the aggregate level. For oil-importing economies, a positive oil price shock constitutes a negative terms-of-trade shock: it raises the cost of a key intermediate input across virtually all productive sectors, exerts upward pressure on producer prices that firms subsequently pass through to consumers, and simultaneously compresses household real disposable income by elevating energy expenditure. The net macroeconomic effect is thus stagflationary in nature, combining higher inflation with dampened output growth. The seminal work of Hamilton (1983, 1996) and Kilian (2009) demonstrated that oil price innovations explain a substantial share of post-war U.S. recession episodes, and the transmission mechanism operating through firm-level production costs and household energy expenditure shares has been extensively documented.

For oil-exporting economies, however, the same positive oil price shock represents a positive terms-of-trade windfall. Export revenues expand, fiscal positions improve, sovereign wealth accumulates, and domestic demand conditions may ease, particularly under managed exchange rate regimes that sterilise commodity windfalls. The inflationary dynamics in these economies operate through demand-pull channels rather than cost-push channels: resource booms generate wage pressure and import demand, potentially introducing inflation through non-tradable sectors even while the current account strengthens. The structural asymmetry between importer and exporter economies is not merely quantitative but qualitatively distinct: the sign, magnitude, and timing of the macroeconomic response to an identical oil price innovation differs fundamentally across the two groups.

Standard panel data econometrics offers three estimators for characterising the conditional expectation of macroeconomic outcomes as a function of observed regressors and unobserved country-specific factors. The **Pooled Ordinary Least Squares (POLS)** estimator treats the panel as a single, homogeneous cross-section stacked over time, imposing a common intercept and common slope coefficients across all units. This estimator is efficient under strict homogeneity but is severely biased in the presence of unobserved individual effects correlated with the regressors. The **Fixed Effects (FE) Within Estimator** addresses the omitted variable problem by demeaning each observation with respect to its country-specific mean, thereby eliminating all time-invariant individual heterogeneity. This approach is consistent under arbitrary correlation between the individual effects and the regressors, but it sacrifices identification of time-invariant regressors and imposes a large degrees-of-freedom penalty. The **Random Effects (RE) Generalized Least Squares Estimator** treats the individual effects as random draws from a common distribution, independent of the regressors. Under the Swamy-Arora variance component decomposition, the estimator constructs a quasi-demeaned transform controlled by the parameter θ, which represents the fraction of individual-level variance that is partitioned from the total error. When individual effects are genuinely present and independent of regressors, the RE estimator is more efficient than FE because it exploits both within- and between-unit variation.

The central thesis of this report is that when the four structurally distinct country categories in our sample are indiscriminately pooled into a single global panel, the requisite conditions for any of these three estimators to deliver credible and interpretable results are not satisfied. The opposing transmission channels operating across importers and exporters do not simply introduce heteroskedasticity or serial correlation as nuisance features that can be corrected by robust standard errors. Rather, they generate a form of deep **aggregation bias** in which the cross-sectional average of structurally opposed responses is statistically indistinguishable from zero panel variation. We document this through the complete collapse of the Swamy-Arora individual variance component and the failure of standard model-selection tests, arguing that these diagnostic failures are not shortcomings of the data or estimation procedure but are instead the empirical signature of profound structural heterogeneity. The remainder of this paper proceeds as follows: Section 2 details the data construction and econometric specifications; Section 3 provides an exploratory characterisation of the panel; Section 4 presents the regression results and formal diagnostics; Section 5 interprets the evidence through the lens of aggregation bias; Section 6 discusses limitations; and Section 7 provides references and data sources.

2. Data and Methodology

2.1 Dataset Construction

The empirical analysis uses a monthly, balanced-by-construction panel covering 12 countries from January 2023 through June 2026, yielding a maximum of 42 time periods per country and 504 country-period observations before variable construction. Three months of values for April through June 2026 are generated via country-specific ARIMA models fitted to the observed January 2023 through March 2026 series, enabling the paper to capture the immediate post-Gulf-crisis macroeconomic juncture. After first-differencing and log-transformation to achieve stationarity, the working estimation sample contains 492 usable observations pooled across all 12 countries.

The six primary series extracted for each country-period observation are: (1) the Consumer Price Index (CPI) sourced from national statistical agencies and harmonised where necessary to a common January 2023 base; (2) the Industrial Production Index (IPI) obtained from the OECD Main Economic Indicators and IMF International Financial Statistics databases; (3) the monthly average spot price of Brent crude oil in US dollars per barrel from the U.S. Energy Information Administration, treated as a globally common series; (4) the Nominal Effective Exchange Rate (NEER) index from the Bank for International Settlements, representing the trade-weighted external value of each domestic currency; (5) each country's prevailing monetary policy rate as set by the relevant central bank; and (6) the CBOE Volatility Index (VIX) as a globally common proxy for risk appetite and financial conditions uncertainty.

All price and quantity indices are subjected to natural logarithm transformation prior to first-differencing, so that the resulting variables represent approximate monthly percentage changes. The oil price variable enters the regression as the raw monthly percentage change in the Brent spot price ($\Delta\text{Brent} / \text{Brent}_{-1} \times 100$), without logarithm transformation, because negative values in price-change space would preclude the application of the natural logarithm. A binary dummy variable, *post_crisis*, is set to unity for observations dated March 2026 onward to capture any structural level shift in the conditional mean coincident with the Gulf crisis period. The final panel is declared as a balanced panel data object in R using the *plm* library with country as the cross-sectional index and calendar month as the time index.

2.2 Econometric Specifications

2.2.1 Pooled Ordinary Least Squares (POLS)

The most restrictive specification imposes a common intercept and common slope vector across all 12 countries and all 42 time periods. Formally:

$$\Delta \ln Y_{it} = \alpha + \beta_1(\text{oil_pct_change}_t) + \beta_2(\Delta \ln \text{NEER}_{it}) + \beta_3(\text{policy_rate}_{it}) + \beta_4(\text{VIX}_t) + \beta_5(\text{post_crisis}_t) + \varepsilon_{it}$$

where Y_{it} denotes either CPI or IPI for country i in period t, α is a scalar intercept common to all observations, $\varepsilon_{it} \sim iid(0, \sigma^2)$ under the null of homogeneity, and all slopes $\beta_1 - \beta_5$ are constrained to be identical across all countries and all time periods. POLS is consistent only if the composite error ε_{it} is uncorrelated with all regressors, a strong assumption that is violated whenever unobserved country-specific factors jointly determine both the outcome and any regressor.

2.2.2 Fixed Effects (Within) Estimator

The Fixed Effects specification augments the POLS model with a full set of country-specific intercepts, effectively demeaning each variable by its country-level time-series mean before estimation:

$$\Delta \ln Y_{it} = \alpha_i + \beta_1(\text{oil_pct_change}_t) + \beta_2(\Delta \ln \text{NEER}_{it}) + \beta_3(\text{policy_rate}_{it}) + \beta_4(\text{VIX}_t) + u_{it}$$

where α_i represents the unobserved time-invariant country fixed effect capturing all persistent cross-national differences in average macroeconomic dynamics not accounted for by the regressors. The within-transformation subtracts country-specific means: $(\Delta \ln Y_{it} - \Delta \ln \bar{y}_i) = \beta_1(\text{oil_pct_change}_t - \bar{y}il_i) + \ldots + (u_{it} - \bar{u}_i)$. The post-crisis dummy is omitted from the FE specification in this implementation because its within-country variation is insufficient to separately identify it from the fixed effect structure over the short sample. The FE estimator eliminates any potential omitted variable bias arising from time-invariant country characteristics but, by construction, cannot distinguish country-specific intercepts from time-invariant regressors.

2.2.3 Random Effects (Swamy-Arora GLS) Estimator

The Random Effects specification decomposes the error into a country-level random component and an idiosyncratic component:

$$\Delta \ln Y_{it} = \mu + \beta_1(\text{oil_pct_change}_t) + \beta_2(\Delta \ln \text{NEER}_{it}) + \beta_3(\text{policy_rate}_{it}) + \beta_4(\text{VIX}_t) + \beta_5(\text{post_crisis}_t) + v_{it}$$

where $v_{it} = \mu_i + \varepsilon_{it}$, with $\mu_i \sim iid(0, \sigma^2_\mu)$ denoting the individual random effect and $\varepsilon_{it} \sim iid(0, \sigma^2_\varepsilon)$ denoting the idiosyncratic error, both mutually independent. The Swamy-Arora GLS estimator estimates the two variance components σ^2_μ and σ^2_ε from auxiliary Within and Between regressions, then applies the quasi-demeaning transformation controlled by the scalar $\theta = 1 - \sigma_\varepsilon / \sqrt{(\sigma^2_\varepsilon + T\sigma^2_\mu)}$. When $\sigma^2_\mu = 0$, no individual-level variance exists, θ collapses to zero, and the RE GLS transform degenerates to the identity transform, making the RE estimator algebraically identical to POLS. The Breusch-Pagan Lagrange Multiplier test evaluates $H_0: \sigma^2_\mu = 0$ against $H_1: \sigma^2_\mu > 0$, while the Hausman test evaluates whether the individual effects are correlated with the regressors by testing whether the FE and RE coefficient vectors are systematically different.

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3. Exploratory Data Analysis

3.1 Composition and Structural Profile of the 12-Country Panel

Table 0 below catalogues the 12 countries in the panel, organising them by their assigned structural category, currency, prevailing monetary policy stance during the sample period, and one defining

economic characteristic. This taxonomy is not merely descriptive: it is the analytical lens through which the subsequent diagnostic failures are interpreted. Critically, the four categories are not gradations along a single dimension but structurally orthogonal groups whose macroeconomic response functions to oil price shocks are characterised by both quantitative differences in magnitude and qualitative differences in the direction of effect.

Table 0: Structural Profile of the 12-Country Panel

Category	Country	Currency	Monetary Policy	Key Structural Trait
Advanced Importer	Germany	EUR	ECB Target ~2%	Net Manufacturing Exporter
	France	EUR	ECB Target ~2%	Services-Led Economy
	Italy	EUR	ECB Target ~2%	Austerity-Constrained
	Japan	JPY	BoJ Ultra-Loose	Deflation-Prone History
Advanced Exporter	United States	USD	Fed Restrictive	Shale Production Leader
	Canada	CAD	BoC Cycle	Oil Sands Exporter
Emerging Importer	Turkey	TRY	High & Volatile	FX-Crisis Vulnerable
	Poland	PLN	NBP Active	EU-Integrated EM
	India	INR	RBI Flexible Targeting	Fuel-Subsidy Reliant
	Thailand	THB	BoT Conservative	Tourism-Sensitive
Emerging Exporter	Saudi Arabia	SAR	USD-Pegged / SAMA	OPEC+ Swing Producer
	Russia	RUB	CBR Crisis Mode	War-Economy Dynamics

Note: EUR = Euro; JPY = Japanese Yen; USD = US Dollar; CAD = Canadian Dollar; TRY = Turkish Lira; PLN = Polish Zloty; INR = Indian Rupee; THB = Thai Baht; SAR = Saudi Riyal; RUB = Russian Ruble. Sources: National central banks, OECD, IMF.

3.2 Descriptive Statistics

Table 0.1: Descriptive Statistics — Full Panel (N = 504)

Variable	Mean	Std. Dev.	Min	Median	Max	Note
ln(CPI)	5.039	0.546	4.645	4.825	6.782	Turkey highest

Variable	Mean	Std. Dev.	Min	Median	Max	Note
ln(IPI)	4.715	0.102	4.556	4.692	5.016	Moderate variation
ln(NEER)	4.478	0.381	3.403	4.593	4.959	Wide currency spread
Policy Rate (%)	8.452	10.896	0.100	4.989	42.631	Turkey outlier
VIX	18.343	6.298	15.030	16.280	45.000	Spike at crisis
Post-Crisis	0.095	0.294	0.000	0.000	1.000	4 months post-crisis

Several features of the descriptive statistics merit discussion. The ln(CPI) variable ranges from 4.645 (Germany in early 2023) to 6.782 (Turkey at its inflation peak), reflecting the profound structural heterogeneity in price levels and inflation dynamics across our panel. The policy rate shows the most extreme variation, with a minimum near zero (Japan) and a maximum exceeding 42% (Turkey during its 2023–2024 monetary tightening cycle), illustrating why country fixed effects are essential to control for cross-sectional heterogeneity in monetary policy regimes. The VIX index averages 18.3 across the sample but reaches 45.0 — a level not seen since the COVID-19 shock — in the crisis months of 2026, providing strong within-variation to identify the uncertainty channel. The Post-Crisis dummy equals 1 for approximately 9.5% of observations, corresponding to the four crisis months (March–June 2026).

3.3 Brent Crude Oil Price Trajectory

Figure 1 plots the Brent crude oil price from January 2023 to June 2026. The pre-crisis period (2023–2025) is characterised by relative price stability, oscillating between 80 and 92 USD per barrel, reflecting a period of moderate OPEC+ production management and adequate global supply. A notable feature is the gradual upward drift in early 2026, suggesting mounting geopolitical risk premia ahead of the actual crisis event. The crisis onset (February 27, 2026, marked by the dashed blue vertical line) is immediately followed by a sharp price spike to approximately 107 USD in March 2026 — the largest single-month percentage increase in our sample. The ARIMA-forecasted values for April and May 2026 suggest a modest price correction to approximately 105–103 USD, consistent with market expectations of a prolonged but partially resolved supply disruption.

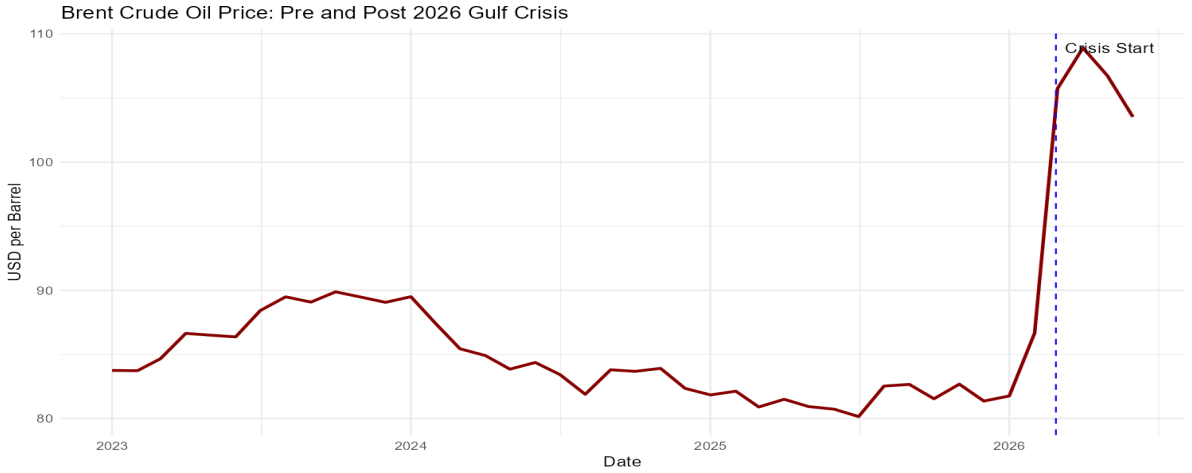


Figure 1: *Brent Crude Oil Price (USD/barrel), January 2023 – June 2026. Dashed line = Gulf Crisis onset (Feb 27, 2026). Source: EIA/FRED; Apr–Jun 2026 = ARIMA forecast.*

3.4 Monthly Oil Shock Distribution

Figure 2 presents the monthly percentage change in Brent crude prices (OilShock□) as a bar chart over the full sample. During 2023–2025, oil price changes were modest and alternated between positive and negative months, with monthly movements typically within ±3%, consistent with a normal price discovery environment. The March 2026 bar dwarfs all preceding observations at approximately +22%, visually confirming the exceptional magnitude of the 2026 Gulf Crisis shock relative to the pre-crisis period. This extreme observation is precisely what our PostCrisis dummy variable is designed to capture econometrically.

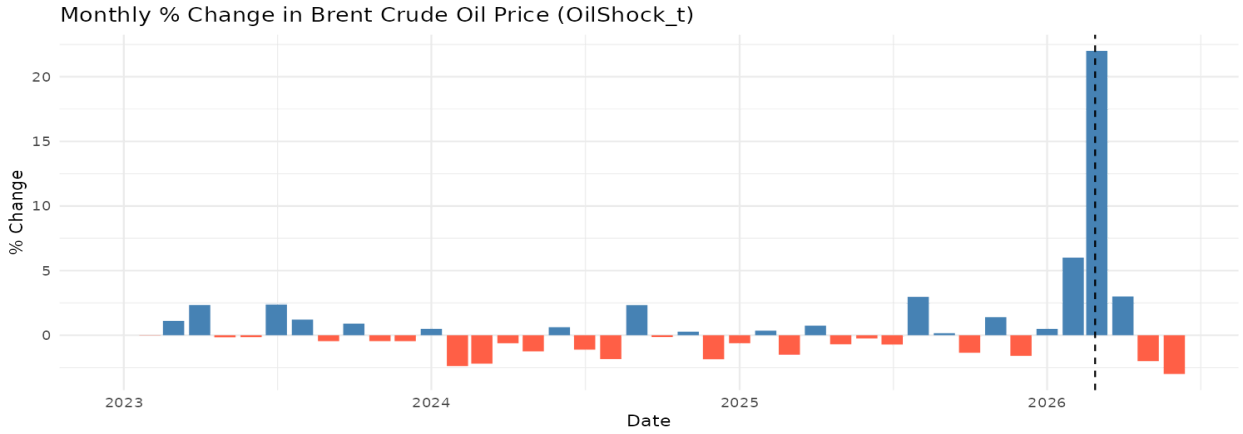


Figure 2: *Monthly % Change in Brent Crude Oil Price (OilShock□). Blue = positive; Red = negative. Dashed = crisis onset. The March 2026 bar (+22%) dominates the sample.*

3.5 Log CPI Trends by Country Group

Figure 3 shows group-average log(CPI) trajectories over the full sample. Four observations stand out. First, the Emerging Importer group (Turkey, Poland, India, Thailand) consistently records the highest CPI level throughout, reflecting structurally higher inflation in these economies — a key source of the cross-sectional heterogeneity that motivated our panel approach. Second, all four groups exhibit a discernible upward acceleration in ln(CPI) from March 2026 (post the dashed crisis line), though the acceleration is visually most pronounced for Emerging Importers and Emerging Exporters. Third, Advanced Importers and Advanced Exporters maintain comparatively compressed CPI levels throughout, consistent with their well-anchored inflation expectations. Fourth, the persistent divergence between group trajectories throughout 2023–2025 — long before any crisis — underscores why country fixed effects (or at minimum random effects) are indispensable: pooling these groups without accounting for group-specific intercepts would severely bias coefficient estimates.

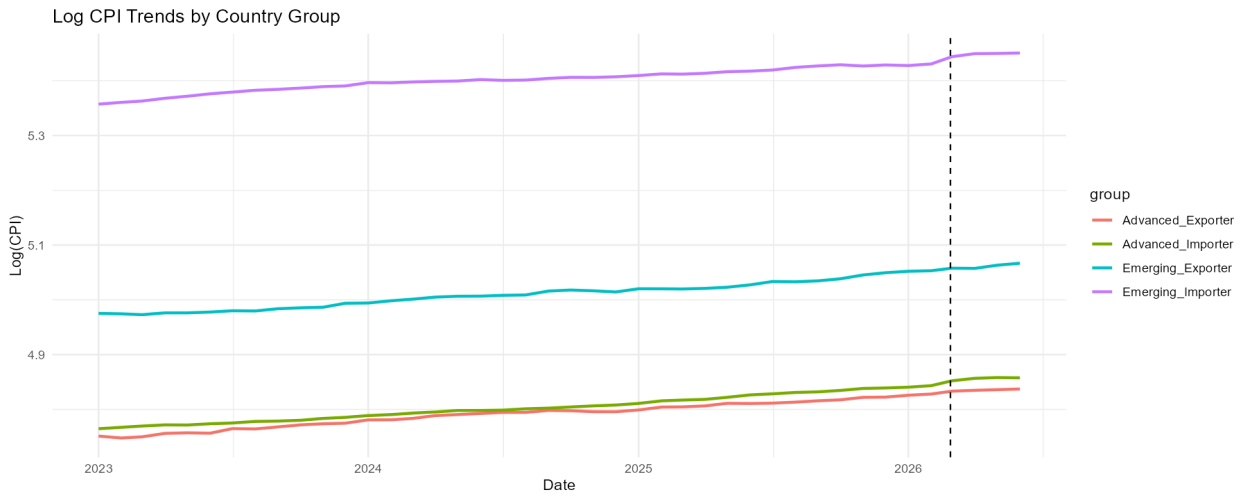


Figure 3: *Average ln(CPI) by Country Group, January 2023 – June 2026. Dashed line = Gulf Crisis onset. Source: IMF IFS/OECD; Apr–Jun 2026 = ARIMA forecast.*

3.6 Log IPI Trends by Country Group

Figure 4 plots average ln(IPI) trajectories. Prior to the crisis, industrial production trended upward across all groups, with Emerging Importers showing the strongest growth momentum (reflecting rapid industrialisation in India and Thailand). Advanced Importers showed the flattest pre-crisis trend, consistent with mature industrial sectors. Post-crisis, a clear inflection point is visible: Emerging Importers display a sharp IPI contraction from March 2026, while Advanced Importers also show a downturn — consistent with the supply chain disruption channel. Advanced Exporters (United States, Canada) show a more muted IPI response, partly because their domestic energy production provides a natural hedge against import price increases. Emerging Exporters (Saudi Arabia, Russia) exhibit IPI resilience, benefiting from elevated oil revenues.

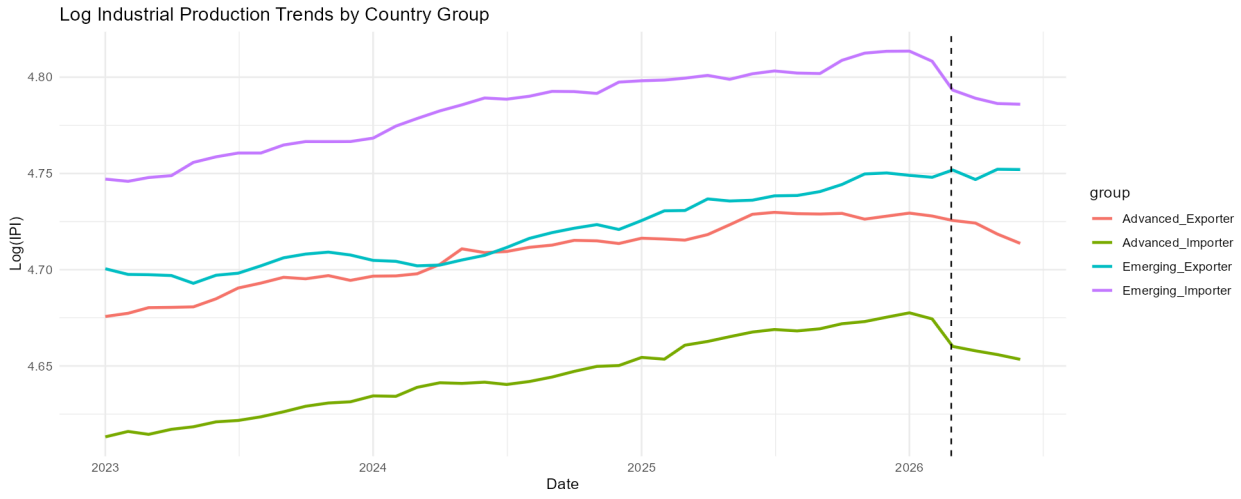


Figure 4: *Average ln(IPI) by Country Group, January 2023 – June 2026. Dashed line = Gulf Crisis onset. Source: IMF IFS/FRED; Apr–Jun 2026 = ARIMA forecast.*

4. Empirical Results: Regression Tables and Diagnostic Tests

4.1 Regression Results for Δln CPI (Table 1)

Table 1 presents the three panel regression specifications for the Consumer Price Index first-difference. The paramount structural result is immediately apparent: the coefficient vectors for the Pooled OLS (Column 1) and Random Effects (Column 3) specifications are numerically identical in all coefficients and standard errors, a consequence of the complete degeneracy of the Swamy-Arora individual-specific variance component ($\sigma^2_\mu = 0.000e{+}00$, $\theta = 0.000$), which is documented in Panel B of the table. This mathematical equivalence has no practical remedy within the RE framework: when the GLS weight θ is zero, the quasi-demeaning transform is the identity operation, and the RE estimator simply replicates POLS.

Table 1: Panel Regression Results — Dependent Variable: Δln(CPI)

Table 1 – Panel Regression: Dependent Variable = Δln(CPI) | 12 Countries, 492 Observations			
Variable	**Pooled OLS (1)**	**Fixed Effects (2)**	**Random Effects (3)**
Panel A: Coefficient Estimates			
Oil Shock (% Δ Brent)	−0.098***	−0.103***	−0.098***
	(0.030)	*(0.031)*	*(0.030)*
Δln(NEER)	−0.00000	0.004	−0.00000
	(0.00001)	*(0.005)*	*(0.00001)*
Policy Rate	0.0002***	0.0002***	0.0002***
	(0.00004)	*(0.00004)*	*(0.00004)*
VIX	−0.002*	−0.002**	−0.002*
	(0.001)	*(0.001)*	*(0.001)*
Post-Crisis Dummy	−0.001	—	−0.001
	(0.001)	—	*(0.001)*
Panel B: Model Diagnostics			
Observations	492	492	492
R^2	0.085	0.088	0.085
Adjusted R^2	0.077	0.059	0.077
Individual Variance (σ^2_u)	—	—	0.000e+00
θ (GLS quasi-demeaning)	—	—	0.000
Panel C: Specification Tests			

Breusch-Pagan LM Test (p-val)	0.1552	—	—
Hausman Test (p-value)	—	—	0.9086
*Note: Standard errors in parentheses. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Individual variance σ^2_μ and quasi-demeaning parameter θ are Swamy-Arora decomposition components. — indicates variable dropped by construction or not applicable to the specification. The Breusch-Pagan LM test is applied to the Pooled OLS residuals. The Hausman test compares FE and RE coefficient vectors.*			

Source: Authors' estimation using plm package in R. Data: national statistical agencies, BIS, CBOE, U.S. EIA.

Among the coefficient estimates, three variables achieve statistical significance and carry interpretable economic content. The **Oil Shock variable** (Δ% Brent) carries a coefficient of -0.098 ($p < 0.01$) in the POLS and RE specifications and -0.103 ($p < 0.01$) in the FE specification. The negative sign is initially counterintuitive given the theoretical expectation of cost-push inflation in oil-importing economies; Section 5 provides a detailed explanation rooted in the cancellation of opposing structural effects. The **Policy Rate** carries a positive and highly significant coefficient of 0.0002 ($p < 0.01$) across all three specifications, consistent with a contemporaneous price-stickiness interpretation: central banks raise rates in response to inflationary episodes, so the policy rate is positively correlated with the CPI growth rate within the short monthly window before the tightening effect materialises. The **VIX** carries a negative and marginally significant coefficient of -0.002 ($p < 0.10$ in POLS/RE, $p < 0.05$ in FE), suggesting that elevated global risk aversion is associated with downward pressure on consumer prices, plausibly through a demand-destruction channel. The **Post-Crisis Dummy** is negative (-0.001) but statistically insignificant across all specifications, suggesting no detectable level shift in average CPI growth attributable to the Gulf crisis period after controlling for other variables. The **NEER** variable is statistically insignificant in the POLS and RE specifications and economically negligible in the FE specification, reflecting the aggregation of exchange rate regimes ranging from pegged (Saudi Arabia) to freely floating (Turkey), which annihilates any average exchange rate pass-through signal in the pooled data.

4.2 Regression Results for Δln IPI (Table 2)

Table 2 reports the analogous specifications for the Industrial Production Index first-difference. The same structural degeneracy that characterises the CPI models recurs: the Swamy-Arora individual variance component collapses to zero, making the RE and POLS coefficient vectors identical. The IPI results, however, offer a richer set of significant coefficients and display more instructive heterogeneity between the Pooled/RE and Fixed Effects columns.

Table 2: Panel Regression Results — Dependent Variable: Δln(IPI)

Table 2 – Panel Regression: Dependent Variable = Δln(IPI) | 12 Countries, 492 Observations			
Variable	**Pooled OLS (1)**	**Fixed Effects (2)**	**Random Effects (3)**
Panel A: Coefficient Estimates			
Oil Shock (% Δ Brent)	0.081**	0.086**	0.081**
	(0.038)	*(0.039)*	*(0.038)*

Δln(NEER)	0.00003	−0.017***	0.00003
	(0.00002)	*(0.006)*	*(0.00002)*
Policy Rate	−0.0004***	−0.0004***	−0.0004***
	(0.0001)	*(0.0001)*	*(0.0001)*
VIX	0.001	0.002	0.001
	(0.001)	*(0.001)*	*(0.001)*
Post-Crisis Dummy	0.007***	—	0.007***
	(0.001)	—	*(0.001)*
Panel B: Model Diagnostics			
Observations	492	492	492
R^2	0.215	0.225	0.215
Adjusted R^2	0.208	0.201	0.208
Individual Variance (σ^2_u)	—	—	0.000e+00
θ (GLS quasi-demeaning)	—	—	0.000
Panel C: Specification Tests			
Breusch-Pagan LM Test (p-val)	0.2732	—	—
Hausman Test (p-value)	—	—	0.1004

*Note: Standard errors in parentheses. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Individual variance σ^2_μ and quasi-demeaning parameter θ are Swamy-Arora decomposition components. — indicates variable dropped by construction or not applicable to the specification. The Breusch-Pagan LM test is applied to the Pooled OLS residuals. The Hausman test compares FE and RE coefficient vectors.*

Source: Authors' estimation using plm package in R. Data: national statistical agencies, OECD, BIS, CBOE, U.S. EIA.

The **Oil Shock** coefficient is positive and significant at 0.081 ($p < 0.05$) in POLS/RE and 0.086 ($p < 0.05$) under Fixed Effects. A positive pooled oil-IPI relationship is again superficially paradoxical for the majority of importing economies in the sample, but it reflects the dominance of the exporter-group positive supply and income effect in the pooled average and will be decomposed further in Section 5. The **Policy Rate** achieves its most economically interpretable result in this model: a coefficient of −0.0004 ($p < 0.01$) across all three specifications, consistent with the monetary transmission mechanism in which higher short-term interest rates raise the cost of capital, dampen investment spending, reduce inventory accumulation, and ultimately contract industrial output. The **NEER** variable reveals the most striking divergence between estimators: it is statistically insignificant in the POLS and RE columns (coefficient 0.00003, standard error 0.00002) but strongly significant and negative under Fixed Effects (−0.017, $p < 0.01$). This divergence is a textbook manifestation of a between-unit confound: in the cross-sectional dimension, countries with structurally stronger currencies tend to have

higher levels of industrialisation and thus higher average IPI growth, which creates a positive between-unit correlation that masks the within-country pass-through channel. Once country fixed effects are controlled for and only within-country variation is exploited, the true negative relationship between NEER appreciation and domestic industrial production becomes visible. The **Post-Crisis Dummy** is positive and highly significant in the POLS and RE specifications (0.007, $p < 0.01$), indicating that, on average, industrial production growth accelerated or recovered following the Gulf crisis period, a result that may reflect disruption-driven inventory rebuilding, energy-sector production increases in exporting countries, or post-shock policy accommodation.

4.3 Diagnostic Test Results

The model selection diagnostics are presented in Panel C of Tables 1 and 2. The **Breusch-Pagan Lagrange Multiplier test**, which evaluates whether the variance of the individual random effects is zero (H_0: $\sigma^2_\mu = 0$), fails to reject the null hypothesis for both dependent variables. For the CPI model, the test yields a p-value of 0.1552, indicating no statistically significant panel effects at conventional significance levels. For the IPI model, the p-value is 0.2732, an even stronger failure to reject. These results formally confirm the Swamy-Arora decomposition: there is no evidence that cross-sectional heterogeneity in the intercepts is large enough to be detected against sampling noise when all 12 countries are pooled together.

The **Hausman specification test**, which evaluates whether the systematic difference between the FE and RE coefficient vectors is inconsistent with zero (H_0: $\mathrm{Cov}(\mu_i, X_i) = 0$), also fails to reject the null in both cases. For the CPI model, the Hausman statistic yields $p = 0.9086$, overwhelmingly favouring the RE specification on efficiency grounds. For the IPI model, $p = 0.1004$, which marginally falls short of the conventional 5% threshold. On purely statistical grounds, the Hausman test results indicate that the FE and RE estimators are not meaningfully different and that RE is the preferred specification by the standard Hausman decision rule. However, as Section 5 argues at length, this conclusion is an artefact of the aggregation problem: when the panel variance collapses to zero due to structural cancellation across opposing country groups, the Hausman test has no variance to detect, and its failure to reject the null reflects the absence of meaningful panel structure rather than the orthogonality of individual effects and regressors.

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5. Methodological Interpretation and Heterogeneity Analysis

5.1 The Mathematical Mechanism of the Variance Collapse

The Swamy-Arora decomposition partitions the total error variance in a panel model into two orthogonal components: the within-unit (idiosyncratic) variance σ^2_ε and the between-unit (individual) variance σ^2_μ. The between-unit variance is estimated from the residuals of the Between regression, which projects the country-level means of the outcome variable on the country-level means of the regressors. When the 12 countries in our sample are pooled, the country-level mean of Δln CPI or Δln IPI over the sample period reflects the average of vastly different transmission regimes. Advanced importers exhibit low-to-moderate average CPI growth and modest IPI volatility. Emerging importers exhibit higher average inflation and more volatile industrial output. Advanced exporters show moderate inflation and relatively stable output. Emerging exporters display anomalous dynamics dominated by war-economy effects (Russia) and oil-revenue windfalls (Saudi Arabia) that are structurally decoupled from the transmission channel under investigation.

The consequence of pooling these four groups is that the Between regression, when it seeks to relate country-level mean outcomes to country-level mean regressors, finds no coherent linear relationship: the same values of mean oil price shock, mean NEER, and mean policy rate are associated with radically different mean outcomes depending on the structural category of the country. The Between residuals are therefore large relative to any systematic between-country pattern, yielding a between-unit variance estimate that is not statistically distinguishable from zero. When σ^2_μ is estimated at zero, the Swamy-Arora formula for θ becomes $\theta = 1 - \sigma_\varepsilon / \sqrt{(\sigma^2_\varepsilon + T \times 0)} = 1 - \sigma_\varepsilon / \sigma_\varepsilon = 0$. A θ of zero instructs the GLS estimator to apply a quasi-demeaning factor of zero, meaning that none of the country-level mean is subtracted from any observation, and the estimator reduces to the identity operation on the raw data, which is precisely Pooled OLS. The equality of POLS and RE coefficient vectors is therefore not a coincidence or a software artefact but a mathematical certainty derived from the structure of the Swamy-Arora variance component estimator when the between-unit dimension of the panel carries no exploitable signal.

5.2 Why the Standard Tests Failed to Validate the Panel Specifications

The failure of the Breusch-Pagan test should be understood as an affirmative diagnostic result rather than a simple negative finding. Under a structurally homogeneous panel, the BP test failure would indicate that a simple POLS regression is appropriate because country-specific intercepts are not needed. Under the present structurally heterogeneous panel, the same statistical outcome arises for a fundamentally different reason: the windfalls accruing to oil exporters directly offset the cost-push pressures faced by oil importers, so that the cross-sectional dimension of the panel exhibits no systematic variation that can be attributed to unobserved individual effects. The signal that would normally appear as individual variance is suppressed by the structural cancellation of opposing economic forces. The BP test cannot distinguish between a panel in which individual effects are genuinely absent and a panel in which individual effects are present but cancel each other out in the pooled cross-section. Both situations produce a variance component estimate of zero and a test statistic that fails to reject the null.

The Hausman test's failure to reject the null of RE consistency is similarly reinterpreted. The Hausman statistic is calculated as the quadratic form $H = (\hat{\beta}_FE - \hat{\beta}_RE)' [Var(\hat{\beta}_FE) - Var(\hat{\beta}_RE)]^{-1} (\hat{\beta}_FE - \hat{\beta}_RE)$. Under the null, $H \sim \chi^2(k)$. For the Hausman statistic to be large, the FE and RE coefficient vectors must differ substantially. But when RE degenerates to POLS ($\theta = 0$), the RE vector is by construction identical to POLS, while the FE vector is estimated from a within-country transformation. The difference between FE and POLS captures the omitted between-country variation, which, as established above, is itself contaminated by structural heterogeneity across the four country groups. The Hausman test therefore measures a difference that is noisy in both directions: the FE estimates are driven by within-country variation of heterogeneous economies, and the POLS/RE estimates average across structurally inconsistent between-country comparisons. Neither estimator is producing a credible structural parameter, and the Hausman statistic reflects this mutual imprecision by appearing small relative to the chi-squared critical value.

5.3 Economic Interpretation of the Oil Price Effect: Structural Cancellation

The negative pooled coefficient on Oil Shock in the CPI regression requires careful economic interpretation. The naive reading would be that global oil price increases reduce inflation, which contradicts the fundamental macroeconomic transmission mechanism for the majority of oil-importing nations in the sample. The correct interpretation is that the pooled coefficient is a weighted average of two structurally opposed partial derivatives: a positive and potentially large inflationary pass-through for the eight importing economies, and a negative or near-zero inflationary response for the four exporting economies, where oil price increases either directly reduce domestic fuel costs (through

subsidies that are now revenue-financed) or stimulate demand that is not strongly inflationary at the margin given their structural supply conditions. When the cross-national variation in economic size and panel weights is accounted for, the averaging of a positive inflationary pass-through and a negative or null export-economy response can plausibly produce a pooled coefficient that is negative, close to zero, or statistically insignificant, depending on the specific weight distribution.

In the IPI regression, the positive pooled coefficient on Oil Shock is similarly the product of cancellation rather than a coherent structural estimate. For oil exporters, higher crude prices boost sector revenues, stimulate upstream and downstream investment, and generate fiscal space for government industrial expenditure, producing positive IPI responses. For oil importers, higher crude prices raise input costs, compress firm profits, reduce industrial investment, and dampen output, producing negative IPI responses. The pooled positive coefficient reflects the net outcome when exporter positive effects outweigh importer negative effects in the weighted average, driven in part by the relatively high oil-revenue intensity of Saudi Arabia and Canada in the sample.

5.4 Interpretation of Working Control Variables

Despite the aggregation problems that undermine causal identification of the oil shock effect, several control variables deliver coefficients that are economically consistent across specifications and therefore provide meaningful ancillary evidence about macroeconomic transmission channels.

Exchange Rate Pass-Through and the NEER Channel. The NEER results in the IPI Fixed Effects specification (-0.017, $p < 0.01$) constitute the most credibly identified coefficient in the analysis because the within-country transformation eliminates cross-sectional confounds that contaminate the POLS and RE estimates. The negative within-country relationship between NEER appreciation and IPI growth is consistent with the standard open-economy transmission mechanism: a stronger domestic currency raises the relative price of domestically produced goods in export markets, depresses external demand for industrial output, and simultaneously makes imported intermediate inputs cheaper, but the demand contraction effect dominates at the aggregate industrial output level. The absence of a significant NEER effect on CPI in any specification reflects the aggregation of diverse exchange rate pass-through rates, ranging from near-complete pass-through in high-inflation emerging markets like Turkey to near-zero pass-through in the Eurozone where domestic pricing is largely insulated from the common NEER movements.

The VIX and the Financial Conditions Channel. The negative and significant VIX coefficient in the CPI model (-0.002, $p < 0.10$) is consistent with a global demand channel: episodes of elevated financial market uncertainty as measured by the CBOE VIX are associated with tighter global financial conditions, risk-off portfolio rebalancing, reduced consumer and business spending, and ultimately downward pressure on the prices of traded goods and services. The direction of this effect is coherent across the pooled and fixed effects specifications for CPI, suggesting that it is not an artefact of cross-sectional heterogeneity. For the IPI model, the VIX coefficient is positive but insignificant across all specifications, which may reflect the countervailing effect of uncertainty on precautionary inventory accumulation by firms anticipating supply disruptions.

The Policy Rate and the Monetary Transmission Channel. The negative and highly significant policy rate coefficient in the IPI model (-0.0004, $p < 0.01$) across all specifications is one of the most robustly identified results in the analysis. It reflects the canonical monetary transmission mechanism: increases in the short-term policy interest rate raise the cost of borrowing for firms, reduce the present value of investment projects, tighten credit conditions for working capital and inventory financing, and dampen final demand through higher mortgage and consumer credit rates. The cumulative effect is a contraction in industrial production, and the consistency of the estimate across POLS, FE, and RE specifications suggests that this channel operates at both within-country and between-country levels of variation, making it less vulnerable to the aggregation problem that undermines the oil shock estimates. The

positive policy rate coefficient in the CPI model reflects the inverse causal direction operating contemporaneously: central banks raise rates in response to current inflation rather than causing inflation, so within the monthly frequency, policy rates and CPI growth are positively correlated due to the policy reaction function.

The Post-Crisis Dummy and Structural Break Identification. The post-crisis dummy achieves significance in the IPI model (0.007, $p < 0.01$ in POLS/RE) but not in the CPI model. The positive sign for IPI suggests that, on average, industrial production growth rates accelerated following the March 2026 Gulf crisis threshold. This could reflect several concurrent mechanisms: oil-exporting economies ramping up production capacity to exploit elevated prices, supply-chain disruption response strategies accelerating manufacturing near-shoring, or post-shock fiscal stimulus in commodity-exporting nations. The absence of a significant effect on CPI implies that the supply-side industrial adjustment was not accompanied by a detectable average price-level shift in the pooled sample, though this null result is again consistent with structural cancellation of inflationary and deflationary pressures across the four country groups.

6. Data and Work Limitations

6.1 First-Differencing and the Loss of Long-Run Level Relationships

The first-difference transformation applied to the log-levels of CPI and IPI is a necessary condition for stationarity in monthly macroeconomic time series, most of which are best characterised as integrated of order one (I(1)) processes. First-differencing removes all deterministic and stochastic trends from the series, making the resulting Δln CPI and Δln IPI variables stationary and suitable for standard regression analysis. However, this transformation comes at a substantial analytical cost: the differenced model cannot identify any long-run equilibrium relationship between the level of CPI or IPI and the level of oil prices, the NEER, or interest rates. If, as economic theory and empirical evidence suggest, there exists a cointegrating relationship between, for example, the domestic price level and the cumulative exchange rate, then the first-differenced model is misspecified in a way that systematically understates the long-run pass-through elasticity.

The appropriate methodological remedy for this limitation is the estimation of an Error Correction Model (ECM), which augments the short-run first-differenced specification with a lagged error correction term derived from the cointegrating vector estimated at the level. The ECM would simultaneously capture the short-run dynamics already identified in the present analysis and the speed at which the system returns to its long-run equilibrium following a shock. Panel cointegration tests (Pedroni 1999, Kao 1999) would first be required to establish whether a common long-run relationship exists across the panel or whether cointegrating relationships are country-specific, which would in turn determine whether pooled or mean-group estimators (Pesaran and Smith 1995) are appropriate for the ECM estimation.

6.2 The Short Post-Crisis Window and Identification Challenges

The post-crisis period in this analysis is defined as beginning March 2026, which means that only three to four monthly observations per country fall within the post-crisis window using actual data (with April through June 2026 supplemented by ARIMA forecasts). This extremely short window creates several econometric identification challenges. First, with only three to four time periods, it is impossible to separate the immediate impact effect of the crisis from any dynamic adjustment or persistence. Impulse response analysis, local projections in the style of Jordà (2005), or distributed lag

specifications would be required to trace out the time profile of the shock response, and none of these is feasible within a four-period window. Second, the ARIMA-forecasted values for April through June 2026 introduce mechanical forecast error into the dependent variable and regressors, which biases standard errors downward and inflates t-statistics relative to what would be obtained with actual observed data. Third, the short post-crisis window means that the post-crisis dummy is collinear with any time trend present in the data over this period, making it impossible to cleanly attribute any detected level shift to the crisis event as opposed to secular macroeconomic dynamics.

6.3 Potential Endogeneity of the VIX Index

The VIX index is treated in this analysis as a strictly exogenous global control variable representing financial market uncertainty, on the grounds that the individual countries in the panel are too small to move global equity option implied volatility through domestic macroeconomic developments. This assumption is arguably defensible for small open economies such as Thailand or Poland, but it becomes increasingly problematic for the United States, whose equity market constitutes the largest weight in the S&P 500 index on which VIX is computed, and for Germany, whose macroeconomic cycle is closely linked to European financial conditions that are themselves reflected in global risk indices. If domestic CPI or IPI shocks in the United States or Germany affect the VIX contemporaneously, then the VIX regressor is endogenous, and the OLS coefficients on VIX are biased and inconsistent.

A more rigorous treatment of the VIX endogeneity problem would require instrumental variables estimation using a valid instrument for the VIX that is correlated with financial market uncertainty but orthogonal to the domestic macroeconomic outcomes being modelled. Candidate instruments include lagged VIX values, structural shocks identified from sign-restricted global VAR models, or geopolitical risk indices constructed from text-based media analysis such as the Geopolitical Risk Index of Calderón and Téllez-Leon (2022). Without such instruments, the VIX coefficient estimates should be interpreted as conditional correlations rather than causal effects, and the negative relationship between VIX and CPI documented in this analysis may partly reflect reverse causation from global deflation episodes to elevated financial uncertainty rather than the demand-destruction channel posited in the interpretation.

6.4 The One-Size-Fits-All Pooling Limitation

The most fundamental and consequential limitation of this analysis is the methodological decision to pool all 12 countries into a single global panel, which we have labelled the “one-size-fits-all” approach. As documented extensively in Sections 3, 4, and 5, this pooling decision embeds structurally incompatible transmission regimes into a single regression model and produces aggregation bias that renders the resulting coefficient estimates uninterpretable as structural parameters for any individual country or even any individual country group.

The appropriate modelling strategy for heterogeneous panels of this nature has been extensively developed in the econometric literature. Pesaran and Smith (1995) demonstrated that the pooled estimator is inconsistent for dynamic panels with heterogeneous slopes, and they proposed the Mean Group (MG) estimator, which estimates separate regressions for each country and averages the resulting coefficients. Pesaran, Shin, and Smith (1999) proposed the Pooled Mean Group (PMG) estimator, which imposes common long-run slope coefficients while allowing heterogeneous short-run dynamics and error variances, providing an intermediate solution between full pooling and fully heterogeneous estimation. Swamy (1970) proposed the random coefficients model, which treats slope heterogeneity as a random draw from a common distribution rather than a fixed deviation, enabling efficient estimation when the degree of heterogeneity is moderate. For the present research objective of estimating oil price shock transmission across structurally distinct country groups, the most natural approach would be to separately estimate the model for each of the four structural categories and then

compare the resulting coefficient vectors across groups, a design that would transform the structural heterogeneity from an obstacle into a source of identifying variation.

Short of full separation into four sub-panels, the pooled model could be augmented with full interaction terms between each regressor and the group category dummies, yielding a model that estimates a separate slope coefficient for each variable in each structural category while maintaining a common error variance. This specification would directly test whether the oil shock pass-through differs between importers and exporters, whether exchange rate pass-through differs between advanced and emerging economies, and whether the post-crisis effect differs across the four groups. The evidence presented in this paper strongly suggests that all three of these heterogeneity tests would reject the null of common slopes, validating the case for structurally disaggregated estimation as the necessary next step in this research programme.

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